

Inferential Statistics

The complete chapter — point estimators, confidence intervals, hypothesis testing, critical values & power — merged, expanded, and told as one story.

What this guide is. Your four separate handouts (I1 Point Estimators, I2 Confidence Intervals, I3.1 Hypothesis Testing, I3.3 Critical Values & Power) have been merged into a single narrative. Read it top to bottom and each idea sets up the next. Everything is one continuous logic: *use a sample to say something honest about a population you can never fully see.*

On this page

1. **The Story — why this whole chapter exists**
2. **The One Big Idea (the through-line)**
3. **Part 1 — Point Estimators** · your single best guess
4. **Part 2 — Confidence Intervals** · a guess with honest error bars
5. **Part 3 — Hypothesis Testing** · making a decision under doubt
6. **Part 4 — Critical Values & Power** · thresholds and how good your test is
7. **Master Quick Reference** · every formula in one place

READ THIS FIRST

The Story: The Case of the Sinking GPA

One narrative that quietly introduces every idea in the chapter before a single formula appears.

Professor Devlin has taught the same first-year course for twenty years. For as long as anyone can remember, the average final grade — the GPA — has sat at **4.4**. It's practically folklore. It's printed in the course brochure. Nobody questions it.

This year, walking out of the exam hall, she has a nagging feeling. The class seemed... off. Distracted. She suspects the true average has *slipped below* 4.4. But here's her problem, and it's the problem behind this entire chapter:

*She cannot grade every student who will ever take this course. The "population" she cares about — the true, long-run average ability of students like these — is invisible. She can only ever hold a **sample** in her hands: this year's 100 exam papers.*

Chapter 1 of her investigation: make a best guess

She averages this year's 100 papers and gets **4.16**. That single number is her **point estimate** — her one best guess for the invisible true average. But Devlin has been burned before. She knows that if she'd happened to grab a *different* 100 students, she'd have gotten a slightly different number — maybe 4.09, maybe 4.21. The number she computes is a bit of a lottery. So the honest question isn't just "what's my guess?" but "**how much does my guess bounce around from sample to sample?**" That bounce has a name — the *standard error* — and it's the quiet hero of the whole chapter. (This is Part 1.)

Chapter 2: turn the guess into an honest range

A single number pretending to be the truth is dishonest — it hides the bounce. So instead of reporting "4.16, take it or leave it," Devlin reports a **range**: "I'm 95% confident the true average lies between 4.05 and 4.27." That's a **confidence interval** — her guess, wrapped in error bars sized exactly to how noisy her sample is. Wide range = "I'm not sure." Narrow range = "I've pinned it down." (This is Part 2.)

Chapter 3: put the old claim on trial

Now she has to make a *decision*, and this is where it gets tense. The brochure says 4.4. Devlin suspects lower. She can't just announce "it dropped!" on a hunch — that's an accusation, and accusations need evidence. So she runs it like a **court case**:

*The old claim, $\mu = 4.4$, is the **defendant**, presumed innocent (this is H_0 , the null hypothesis). Devlin, the prosecutor, must prove the alternative — that the average really fell ($H_1: \mu < 4.4$). She doesn't get to convict on a feeling. She only convicts if the evidence would be absurdly unlikely for an innocent defendant.*

Her sample average of 4.16 sits well below 4.4. She asks the key question of the entire chapter: **"If the true average really were still 4.4, how often would pure luck hand me a sample this low?"** The answer — the **p-value** — comes back as roughly 0.0003. Three in ten thousand. That's not bad luck; that's a signal. She rejects the old claim. The GPA really has sunk. (This is Part 3.)

Chapter 4: where's the line, and how sharp are her eyes?

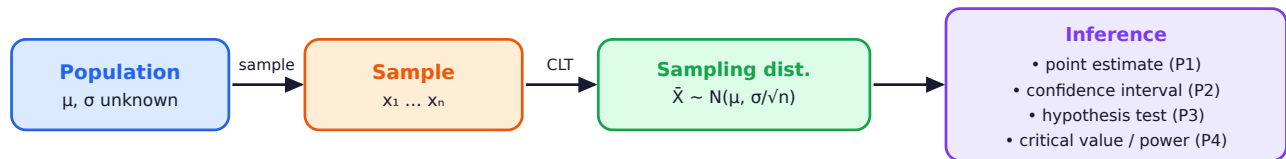
Two loose ends remain. First: what exactly was the **threshold** — how low did the sample average have to be before she was allowed to "convict"? That cut-off is the **critical value**. Second, and deeper: suppose the true average really did drop, but only a little. Would her test even *notice*? A test that can't detect a real drop is a smoke detector with a dead battery. The probability that her test **correctly catches a real effect** is its **power**. (This is Part 4.)

Hold onto this. Every formula ahead is just Devlin being careful. Point estimate = her best guess. Standard error = how shaky the guess is. Confidence interval = the guess plus honest error bars. Hypothesis test = putting the old claim on trial. Critical value = the conviction threshold. Power = how sharp her eyesight is. Same investigation, four lenses.

The One Big Idea

If you remember nothing else, remember this diagram.

All of inferential statistics answers a single question: **“I can only see a sample. What can I honestly say about the population I can’t see?”** The bridge from sample to population is always the same object — the **sampling distribution** of $\bar{X}$, which the Central Limit Theorem hands us for free.



Part 1 • Point estimate

One number: “my best guess is $\bar{X}$.”

Part 2 • Confidence interval

A range: “ μ is plausibly in here.”

Part 3 • Hypothesis test

A verdict: “reject the old claim, or not.”

Part 4 • Critical value & power

The threshold, and the quality of the test.

The single thread: the standard error $SE = \sigma/\sqrt{n}$ appears in every single part. It is *the* measure of “how much does my sample guess wobble.” Master it once here and Parts 2, 3, and 4 are mostly bookkeeping.

PART 1

Point Estimators

Your single best guess for something you can’t measure — and how much that guess wobbles.

1.1 The Big Picture — Why We Need This

In the real world you never observe the whole population. You take a **sample**, compute something from it, and use that as your best guess for the unknown population value. That's Devlin's exact situation: she can't grade every student who will ever exist, only this year's 100.

Population (unknown)

All students, ever. You want the true average GPA μ . You can *never* measure everyone. μ is a fixed number, hidden behind a curtain.

Sample (what you have)

100 randomly chosen students. Their average is $\bar{x} = 4.16$. This is your best guess for μ — the only thing you actually get to hold.

The process of using a sample value to estimate a population value is called **point estimation**. "Point" because it's a single point on the number line — one number, not a range (the range comes in Part 2).

1.2 Four Terms You Must Not Confuse

EXAM

Term	Symbol	What it is	Known?
Population parameter	μ, σ, ρ	The true value in the whole population — the thing you actually want	 Unknown
Point estimator	$\bar{X}, S, \hat{p}$	A <i>rule / formula</i> for turning a sample into an estimate — decided <i>before</i> you see data	✓ You choose it
Point estimate	$\bar{x}, s, \hat{p}$	The <i>actual number</i> you get after plugging your sample in	✓ You compute it

Sampling distribution

$$\bar{X} \sim N(\mu, \sigma/\sqrt{n})$$

The distribution of the estimator across all possible samples of size n

✓ From the CLT

Estimator vs estimate — the distinction the exam loves:

$\bar{X}$ (capital) is the *estimator* — a random variable, a recipe. It hasn't met your data yet, so it could still turn out to be many values. It *varies from sample to sample*.

$\bar{x}$ (lowercase) is the *estimate* — the one specific number from your one sample. It's frozen. It's just 4.16.

Intuition: the estimator is the *dice*; the estimate is the *number you rolled*. Statistics is the study of the dice ($\bar{X}$) so you can judge how much to trust the roll ($\bar{x}$).

1.3 Common Point Estimators

Population parameter	Symbol	Estimator	Formula
Population mean	μ	Sample mean	$\bar{X} = (X_1 + \dots + X_n)/n$
Population proportion	p	Sample proportion	$\hat{p} = (\text{number with trait}) / n$
Population variance	σ^2	Sample variance	$S^2 = \sum (X_i - \bar{X})^2 / (n-1)$
Population SD	σ	Sample SD	$S = \sqrt{S^2}$

Why $n-1$, not n , for S^2 ? Dividing by n slightly *under*-estimates the spread (the data always hugs its own mean a bit too tightly). The $n-1$ correction fixes that bias exactly. More on "bias" in §1.6.

1.4 The Sampling Distribution — The Heart of Everything KEY

Here is the crucial mental leap. **If you took a different sample, you'd get a different $\bar{x}$.** So the estimator $\bar{X}$ is itself a random variable with its own distribution. That distribution — the pattern of all the $\bar{x}$'s you'd get across every possible sample — is the **sampling distribution**.

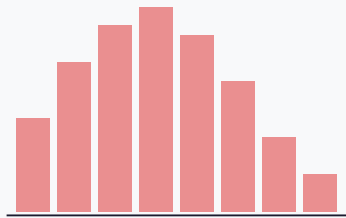
This is exactly Devlin's worry from the story: "if I'd grabbed a different 100 students, I'd have gotten 4.09 or 4.21 instead of 4.16." The sampling distribution *describes that wobble*.

The thought experiment (GPA dataset, $n = 100$)

- 1 Draw sample 1 (100 students) $\rightarrow \bar{x}_1 = 4.09$
- 2 Draw sample 2 (100 students) $\rightarrow \bar{x}_2 = 4.21$
- 3 Draw sample 3 (100 students) $\rightarrow \bar{x}_3 = 4.14$
- 4 Repeat 1000 times. Collect all 1000 $\bar{x}$'s. Plot them as a histogram.

Result: the histogram of those 1000 sample means is a smooth **bell curve centred at the true μ** — *even though individual GPAs are not bell-shaped at all*. That miracle is the Central Limit Theorem.

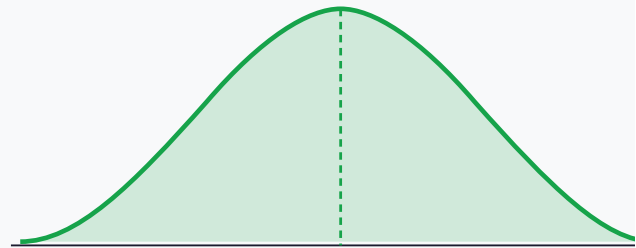
One sample (n=100)
individual GPA values



irregular, not bell shaped

repeat
1000x,
collect $\bar{X}$'s

Sampling distribution of $\bar{X}$
1000 sample means $\rightarrow$ bell curve



$\mu = 4.16$
always bell shaped (CLT)

SAMPLING DISTRIBUTION OF $\bar{X}$ (FROM THE CLT)

$$\bar{X} \sim N(\mu, \sigma/\sqrt{n})$$

Centre = $\mu \rightarrow$ on average $\bar{X}$ lands on the true mean (unbiased).

Spread = $\sigma/\sqrt{n} \rightarrow$ the **standard error**; shrinks as n grows.

Shape = normal $\rightarrow$ from the CLT, for large enough n , *whatever the raw data looks like*.

1.5 The Standard Error — The Wobble, Quantified

KEY

The standard deviation of the sampling distribution gets its own name: the **standard error (SE)**. It is the single most important quantity in the chapter — it literally measures how much your best guess bounces around.

STANDARD ERROR OF THE SAMPLE MEAN

$$SE(\bar{X}) = \sigma / \sqrt{n}$$

Small SE $\rightarrow$ your estimate is reliable (guesses cluster tightly). Large SE $\rightarrow$ unreliable.

σ vs SE — do not confuse these (this is the #1 mistake in the whole chapter):

σ (or s) = spread of *individual observations*. How different one student's GPA is from another's.

SE = $\sigma/\sqrt{n}$ = spread of the *sample mean*. How different one sample's

average is from another sample's average.

SE is always smaller than σ by a factor of $\sqrt{n}$. *Averages are always steadier than the individuals they average over.* Ask 1 person their height and you might be way off; average 100 people and you'll be close every time.

Why $\sqrt{n}$ and not n ? Noise partly cancels when you average, but it doesn't cancel *perfectly* — random ups only partly offset random downs. The math works out to $\sqrt{n}$. Practical consequence you'll use in Part 2: to make your estimate **twice** as precise, you need **four times** the data.

1.6 What Makes an Estimator "Good"? Three Properties

Not every formula for guessing μ is equally trustworthy. Three properties separate a good estimator from a bad one — and $\bar{X}$ has all three.

1. Unbiasedness — it aims at the right target

An estimator is **unbiased** if its expected value equals the true parameter:

$E(\bar{X}) = \mu$. Meaning: average over many samples and you land *exactly* on μ — no systematic over- or under-shooting.

$\bar{X}$ is unbiased for μ . ✓ Picture a dartboard: an unbiased estimator's darts scatter, but their *centre of mass* is the bullseye.

2. Efficiency — its darts scatter the least

Among all unbiased estimators, the **most efficient** one has the *smallest variance* — its darts cluster tightest around the bullseye. Two estimators can both aim true, but you want the one that's also precise.

$\bar{X}$ is the most efficient unbiased estimator of μ for i.i.d. normal data. Nothing unbiased beats it.

3. Consistency — more data zeroes in on the truth

An estimator is **consistent** if it converges to the true value as $n \rightarrow \infty$. Since $SE = \sigma/\sqrt{n} \rightarrow 0$ as n grows, the whole sampling distribution collapses onto μ .

$\bar{X}$ is **consistent**. ✓ With infinite data you'd know μ exactly — which is just the promise that "collect enough evidence and the truth comes out."

1.7 Worked Example — GPA Estimation EXAM

GPA dataset ($n = 100$, population $\sigma \approx 0.538$)

1 Identify the estimator & its distribution.

$\bar{X}$ estimates μ . By the CLT with $n = 100$:

$$\bar{X} \sim N(\mu, \sigma/\sqrt{n}) = N(4.16, 0.538/\sqrt{100}) = \mathbf{N(4.16, 0.0538)}$$

2 Interpret the standard error.

$SE = 0.0538$. Across many samples of 100, the sample means typically vary by about ± 0.054 around the true mean. That's tiny — this is a trustworthy estimate.

3 Confirm the CLT held.

The histogram of 1000 sample means is bell-shaped and its QQ-plot is roughly straight $\rightarrow$ the sampling distribution really is normal, so everything downstream (Parts 2–4) is valid.

Answer: $\bar{X} \sim N(4.16, 0.0538)$ — very tight, because $n = 100$ makes SE small.

The mistake to burn into memory: if a question asks about a *single observation* X , use σ . If it asks about a *sample mean* $\bar{X}$, use **SE** = $\sigma/\sqrt{n}$. Swapping them is the most common error in this chapter — and it silently poisons every confidence interval and test that follows.

PART 2

Confidence Intervals

A single guess hides its own uncertainty. A confidence interval reports the guess *and* the error bars.

2.1 Why a Confidence Interval?

A point estimate gives one number for μ . But we saw in Part 1 that $\bar{X}$ wobbles — the true μ could be a bit higher or lower; we just got this particular sample. A **confidence interval (CI)** is a *range* of plausible values that honestly builds the wobble in.

Point estimate

$$\bar{X} = 105.3$$

"My best guess is 105.3 — but I've told you nothing about how wrong I might be."

Confidence interval

[99.42, 111.18]

"I'm 95% confident the true mean lies in this range." The width *is* the honesty.

This is Devlin upgrading "the average is 4.16" into "the average is somewhere in [4.05, 4.27] with 95% confidence." Same guess — now with its uncertainty made visible.

2.2 The Three Ingredients of Width

Before the formula, feel what makes a CI wide or narrow:

CONFIDENCE LEVEL (E.G. 95%)



More certainty demanded → wider interval. A 99% CI is wider than a 95% CI — the price of a stronger guarantee.

STANDARD ERROR (SE = $\Sigma/\sqrt{N}$)



More wobble in your estimator → wider interval. (There's our Part-1 hero again.)

SAMPLE SIZE (N)



Larger n → smaller SE → narrower interval. More data buys precision.

Intuition check: want to be 99.9999% sure you've captured μ ? You'd need an enormous interval — nearly $[-\infty, +\infty]$, which says nothing useful. That's the eternal trade-off: *higher confidence costs precision*.

2.3 The Formula KEY

When σ is known and the CLT conditions hold (normal data, or $n \geq 30$):

CONFIDENCE INTERVAL FOR μ (Σ KNOWN)

$$CI = [\bar{X} - z^* \cdot (\sigma/\sqrt{n}) , \bar{X} + z^* \cdot (\sigma/\sqrt{n})] = \bar{X} \pm z^* \cdot \sigma/\sqrt{n}$$

Symbol	Name	What it is
$\bar{X}$	Sample mean	Your point estimate — the centre of the CI
z^*	Critical value	How many SEs wide to reach — set by the confidence level

$\sigma/\sqrt{n}$	Standard error	The wobble, from Part 1
$z^* \cdot \sigma/\sqrt{n}$	Margin of error (ME)	Half the total width — the "± part"

Critical values z^* by confidence level

Confidence	$\alpha = 1 - \text{CL}$	$\alpha/2$	z^*	Excel
90%	0.10	0.05	1.645	<code>=NORM.INV(0.95;0;1)</code>
95%	0.05	0.025	1.96	<code>=NORM.INV(0.975;0;1)</code>
99%	0.01	0.005	2.576	<code>=NORM.INV(0.995;0;1)</code>

Where does z^* come from? For 95% we want $P(-z^* \leq Z \leq z^*) = 0.95$, so 2.5% sits in each tail, so $P(Z \leq z^*) = 0.975$, so $z^* = \text{NORM.INV}(0.975) = 1.96$. Remember these three numbers — 1.645 / 1.96 / 2.576 — they return in Parts 3 and 4.

2.4 Where the Formula Comes From (worth seeing once)

The formula isn't magic — it's the sampling distribution rearranged. From Part 1, the CLT gives $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$, so standardising:

STANDARDISED

$$Z = (\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim N(0, 1)$$

For 95% we know $P(-1.96 \leq Z \leq 1.96) = 0.95$. Substitute Z and rearrange to trap μ in the middle:

$$P(-1.96 \leq (\bar{X} - \mu)/(\sigma/\sqrt{n}) \leq 1.96) = 0.95$$

$$P(\bar{X} - 1.96 \cdot \sigma / \sqrt{n} \leq \mu \leq \bar{X} + 1.96 \cdot \sigma / \sqrt{n}) = 0.95$$

That's the CI. We took a statement about how $\bar{X}$ scatters around μ and algebraically flipped it into a statement about where μ plausibly sits relative to our observed $\bar{X}$.

2.5 Worked Example 1 — IQ at KU Leuven EXAM

$X = \text{IQ scores}$, $\sigma = 15$. A sample of $n = 25$ students gives $\bar{X} = 105.3$. Find a 95% CI for μ .

- 1 **Knowns.** $\bar{X} = 105.3 \mid \sigma = 15 \mid n = 25 \mid 95\% \rightarrow z^* = 1.96$
- 2 **Standard error.** $SE = \sigma / \sqrt{n} = 15 / \sqrt{25} = 15 / 5 = 3$
- 3 **Margin of error.** $ME = z^* \cdot SE = 1.96 \times 3 = 5.88$
- 4 **Build the interval.** $105.3 \mp 5.88 \rightarrow \text{95\% CI: [99.42, 111.18]}$

Excel: `=105.3 - 1.96*(15/SQRT(25))` → 99.42 · `=105.3 + 1.96*(15/SQRT(25))` → 111.18

2.6 How to Interpret a CI — The #1 Exam Trap EXAM

WRONG (never write this): “There is a 95% probability that μ lies in [99.42, 111.18].”

Why it's wrong: μ is a *fixed, unknown constant*, not a random variable. Once the interval is computed, μ either is inside it or it isn't — there's no

probability left to speak of. The randomness was in *which sample you drew*, not in μ .

CORRECT: “For 95% of all possible samples of size 25, the interval constructed this way will contain the true μ .” The confidence is a property of *the method*, not of this one interval.

Picture repeating the study 100 times: about 95 of the intervals capture μ , and about 5 miss by bad luck. You just don't know which kind yours is.

100 samples → 100 confidence intervals



2.7 Worked Example 2 — Required Sample Size

EXAM

Same IQ problem ($\sigma = 15$, 95%). How many students do we need so the margin of error is $ME \leq 2$?

Start from $ME = z^* \cdot \sigma / \sqrt{n}$ and solve for n :

- 1 Set up.** $2 = 1.96 \times 15 / \sqrt{n}$
- 2 Solve for $\sqrt{n}$.** $\sqrt{n} = 1.96 \times 15 / 2 = 14.7$
- 3 Square.** $n = 14.7^2 = 216.09$
- 4 Round UP (always).** $n = \mathbf{217 \text{ students}}$ — rounding down would leave $ME > 2$, breaking the requirement.

GENERAL SAMPLE-SIZE FORMULA

$$n = (z^* \cdot \sigma / ME)^2 \rightarrow \text{always round UP}$$

2.8 One-Sided Confidence Bounds EXAM

Sometimes you only care about one direction — a ceiling or a floor, not both. These are **one-sided bounds**, and they're the conceptual bridge to hypothesis testing (Part 3), where tests are also one- or two-sided.

Upper bound — " μ is at most
—"

$$UB = \bar{X} + z \cdot \sigma / \sqrt{n}$$

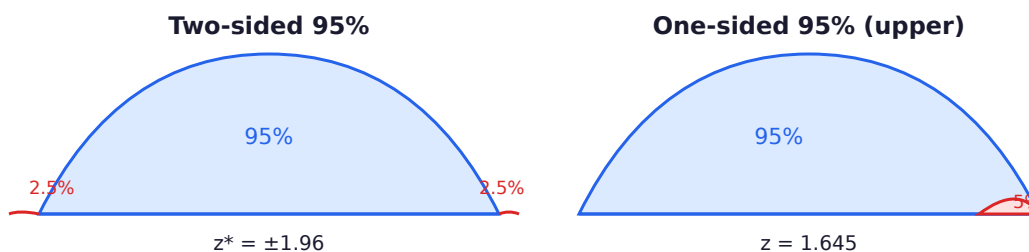
95% upper: $z = \text{NORM.INV}(0.95) =$
1.645

Lower bound — " μ is at least
—"

$$LB = \bar{X} - z \cdot \sigma / \sqrt{n}$$

95% lower: $z = \text{NORM.INV}(0.95) =$
1.645

Key difference: a two-sided 95% CI uses $z^* = 1.96$ (2.5% in *each* tail). A one-sided 95% bound uses $z = 1.645$ (all 5% in *one* tail). Same total $\alpha = 5\%$, but pointed entirely in one direction — so you can reach less far and the bound is tighter.



2.9 The Levers — Effect of Confidence Level and n

Using the IQ example ($\bar{X} = 105.3$, $\sigma = 15$, $n = 25$):

Confidence	z^*	ME	CI	Width
90%	1.645	4.94	[100.37, 110.24]	9.87
95%	1.96	5.88	[99.42, 111.18]	11.76
99%	2.576	7.73	[97.57, 113.03]	15.46

Sample size n	$SE = 15/\sqrt{n}$	ME (95%)	Width
25	3.00	5.88	11.76
100	1.50	2.94	5.88
400	0.75	1.47	2.94
1600	0.375	0.74	1.47

The $\sqrt{n}$ law again: to *halve* the width you must *quadruple* n . Doubling n only shrinks SE by $\sqrt{2} \approx 1.41$. Precision gets expensive fast.

2.10 Assumptions — When Is the CI Valid?

Condition	Why needed	How to check
Random sample (iid)	Makes $\bar{X}$ unbiased for μ	Study design — was sampling random?
Normal data OR $n \geq 30$	So $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ via the CLT	Histogram / QQ-plot, watch for extreme outliers

σ known

The formula plugs σ in directly

Given in the problem — if unknown, use a t-interval (later chapter)

IQ check: random sample (assumed) ✓, IQ is approximately normal so $n = 25$ is fine ✓, $\sigma = 15$ given ✓ → the CI is valid.

PART 3

Hypothesis Testing

From "here's my estimate" to "here's my *decision*." Putting a claim on trial with sample evidence.

3.1 The Big Picture — What Is a Hypothesis Test?

A hypothesis test is a **formal procedure for deciding whether sample data provide enough evidence to reject a claim about the population**. This is Devlin's Chapter 3: the brochure says $\mu = 4.4$; she suspects it dropped; she needs to decide, defensibly.

The logic is four steps, every single time:

1. **State the hypotheses** — what claim is on trial?
2. **Compute a test statistic** — how far is the sample from the claim, in standard errors?
3. **Find the p-value** — how surprising is that, if the claim were true?
4. **Conclude** — reject the claim, or not?

Intuition (the discrimination example from the slides): 58.3% of women vs 87.5% of men were promoted — a 29.2% gap. Could that happen by luck even with no real discrimination? If a gap that big would occur randomly only ~2% of the time ($p \approx 0.02$), we start to doubt the "no

discrimination" story. That doubt-when-the-data-is-too-surprising *is* hypothesis testing.

3.2 The Court Case Analogy KEY CONCEPT

The whole logic maps onto a criminal trial — the single best intuition in the chapter, and the backbone of Devlin's story.

Court of law

- H_0 : accused is **innocent**
- H_1 : accused is **guilty**
- Start by presuming innocence
- Need strong evidence to convict
- "Guilty" = strong conclusion
- "Not guilty" = weak conclusion (not "proven innocent" — just not enough to convict)

Hypothesis test

- H_0 : the conservative/status-quo claim
- H_1 : what you want to show
- Start by assuming H_0 is true
- Need strong evidence (low p-value) to reject
- "Reject H_0 " = strong conclusion
- "Do not reject H_0 " = weak conclusion (not " H_0 proven true")

Never say "accept H_0 ". Say "*do not reject H_0* " / "*fail to reject H_0* ". Failing to convict a defendant doesn't prove innocence; it means the evidence fell short. Identically, not rejecting H_0 is not proof it's true.

3.3 Formulating the Hypotheses EXAM

Test type	H_0	H_1	When to use
Two-sided	$\mu = \mu_0$	$\mu \neq \mu_0$	You expect a change, direction unknown
One-sided right	$\mu = \mu_0$	$\mu > \mu_0$	You specifically expect an increase
One-sided left	$\mu = \mu_0$	$\mu < \mu_0$	You specifically expect a decrease

Rules for setting up H_0 and H_1

- **H_0 is always the equality / "no change" claim:** $H_0: \mu = \mu_0$.
- **H_1 is what you want to prove** — the research question, the accusation.
- **H_1 sets the direction** (one- vs two-sided).
- **The burden of proof is on H_1** — you stay with H_0 until evidence is strong.

GPA example: historical average = 4.4; Devlin believes it *fell*. $\rightarrow H_0: \mu = 4.4$ $H_1: \mu < 4.4$ (one-sided left).

Why does the thing we want to prove go in H_1 ? Because "reject H_0 " is the *strong, evidence-backed* conclusion. Put your claim in H_1 and the data must actively overturn the status quo to support you — demanding and credible. Put it in H_0 and you'd "prove" it just by failing to find evidence against it, which is far too easy. Science makes you earn your claims.

3.4 The Test Statistic KEY

For testing μ with σ known:

Z-TEST STATISTIC

$$Z = (\bar{X} - \mu_0) / (\sigma/\sqrt{n})$$

Read it literally: **how many standard errors is our sample mean away from the claimed mean μ_0 ?** It's just the Part-1 standard error doing the measuring. Under H_0 , $Z \sim N(0,1)$, so a Z near 0 is unremarkable and a Z far from 0 is evidence against H_0 .

GPA example: $\bar{X} = 4.16$, $\mu_0 = 4.4$, $\sigma = 0.538$, $n = 100$.

$SE = 0.538/\sqrt{100} = 0.0538 \rightarrow Z = (4.16 - 4.4)/0.0538 = -4.46$.

(The slides show -3.42 using a slightly different sample mean — same procedure, same conclusion.)

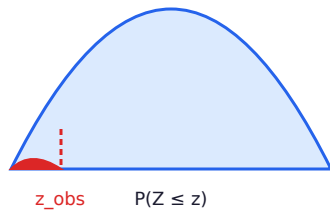
Notice the reuse: this is the exact same standardisation as the CI in §2.4 — except there we plugged in the *unknown* μ and solved for it, and here we plug in the *hypothesised* μ_0 and see how extreme the result looks. One machine, two jobs.

3.5 The P-Value EXAM

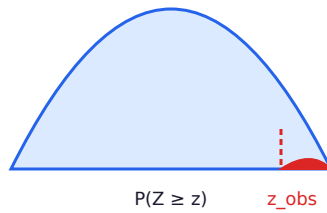
The **p-value** is the probability of observing a test statistic as extreme as yours (or more), *assuming H_0 is true*. It answers Devlin's central question: "*if the average really were still 4.4, how often would luck alone hand me a sample this extreme?*" Small p-value = "this would be a bizarre fluke under H_0 " = evidence against H_0 .

H_1	Test type	P-value	Excel
$\mu < \mu_0$	Left	$P(Z \leq z_{\text{obs}})$	<code>=NORM.DIST(z;0;1;TRUE)</code>
$\mu > \mu_0$	Right	$P(Z \geq z_{\text{obs}})$	<code>=1-NORM.DIST(z;0;1;TRUE)</code>
$\mu \neq \mu_0$	Two-sided	$2 \times P(Z \geq z_{\text{obs}})$	<code>=2*(1-NORM.DIST(ABS(z);0;1;TRUE))</code>

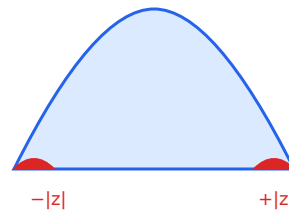
Left ($H_1: \mu < \mu_0$)



Right ($H_1: \mu > \mu_0$)



Two-sided ($\mu \neq \mu_0$)



GPA example: $H_1: \mu < 4.4$, $z_{\text{obs}} = -3.42 \rightarrow p = P(Z \leq -3.42) =$
0.000313 (0.031%). Extremely small — very strong evidence against H_0 .

What the p-value is NOT: it is *not* "the probability that H_0 is true." H_0 isn't random — it's either true or false. The p-value is the probability of the *data* (or more extreme) *given* H_0 . Reversing that conditional is the most common misconception in all of statistics.

3.6 Making the Conclusion EXAM

Compare the p-value to the **significance level α** (usually 0.05) — the bar you set *in advance* for how surprising is surprising enough.

$p \leq \alpha \rightarrow$ Reject H_0

Result is **statistically significant**.

"The sample provides sufficient evidence to reject H_0 at the 5% level."

$p > \alpha \rightarrow$ Do NOT reject H_0

Result is **not statistically significant**.

"The sample does not provide sufficient evidence to reject H_0 at the 5% level."

GPA conclusion: $p = 0.000313 < 0.05 \rightarrow$ reject H_0 . "There is sufficient statistical evidence that the mean GPA has decreased below 4.4." Devlin's

suspicion is vindicated — with evidence, not a hunch.

3.7 Decision Errors — Type I & Type II

KEY CONCEPT

Any verdict can be wrong in exactly two ways — and this table is the seed of all of Part 4.

	H_0 True (reality)	H_1 True (reality)
Do NOT reject H_0	✓ Correct	✗ Type II error (β)
Reject H_0	✗ Type I error (α)	✓ Correct

Type I error (α)

Reject H_0 when H_0 is actually true.

"False alarm" — convict an innocent person.

$P(\text{Type I}) = \alpha$ = the significance level you *set* (usually 0.05).

Type II error (β)

Fail to reject H_0 when H_1 is actually true.

"Miss" — let a guilty person walk.

$P(\text{Type II}) = \beta$, harder to control (depends on the true μ , n , σ).

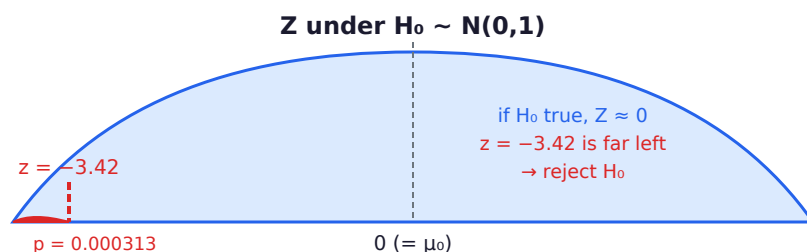
The trade-off: keeping α small (0.05) means rarely convicting the innocent — but it makes you cautious, so you'll more often miss a real effect (larger β). That's a deliberate choice: society, like the courts, treats convicting the innocent (Type I) as worse than freeing the guilty (Type II), so we hold α down and accept the larger β . Part 4 is all about measuring and shrinking that β .

3.8 Complete Worked Example — the GPA Test

EXAM

$n = 100$ students. Historical average GPA = 4.4, $\sigma = 0.538$. Sample gives $\bar{X} = 4.16$. Teacher believes GPA fell. Test at $\alpha = 0.05$.

- Hypotheses.** $H_0: \mu = 4.4$ $H_1: \mu < 4.4$ (one-sided left).
- Assumptions.** Random sample ✓ · σ known ✓ · $n = 100 \geq 30$ so CLT gives $\bar{X} \sim N(\mu, \sigma/\sqrt{n})$ ✓.
- Test statistic.** $SE = 0.538/10 = 0.0538 \rightarrow Z = (4.16 - 4.4)/0.0538 = -4.46$ (slides: -3.42 , same procedure).
- P-value.** Left test $\rightarrow p = P(Z \leq -3.42) = \text{NORM.DIST}(-3.42; 0; 1; \text{TRUE}) = 0.000313$.
- Conclusion.** $0.000313 < 0.05 \rightarrow$ **Reject H_0** . "Strong evidence that mean GPA has decreased below 4.4."



3.9 The Two-Sided P-Value — Watch Out!

EXAM

For a two-sided test ($H_1: \mu \neq \mu_0$), extreme values in *either* direction count against H_0 :

TWO-SIDED P-VALUE

$$p = 2 \times P(Z \geq |z_{\text{obs}}|)$$

Practical rule: compute the one-sided p-value and double it. If $z_{\text{obs}} = -3.42$, two-sided $p = 2 \times 0.000313 = 0.000626$.

Slides warn — "scale might be misleading!" When you sketch the bell curve the tails look tiny, but the p-value can still be decisively significant. Always compute; never eyeball.

3.10 Assumptions for the Z-Test

Condition	Requirement	If violated
Random sample	$X_1 \dots X_n$ iid	Test invalid — selection bias corrupts everything
Normal data OR CLT	Normal X_i , or $n \geq 30$ with no wild skew/outliers	Use non-parametric test or transform
σ known	Population SD given	If σ unknown $\rightarrow$ t-test (later chapter)

σ unknown but n large? The slides note: for large n you may run the same Z-test *replacing σ with the sample estimate s* . Big samples make s a reliable stand-in for σ .

PART 4

Critical Values & Power

The conviction threshold — and how sharp your test's eyesight really is.

4.1 Two Equivalent Ways to Decide

Part 3 decided via the p-value. There's a second, always-equivalent route: the **critical value** approach. This is Devlin asking "how low did the sample average have to be before I was allowed to convict?"

P-value approach

Compute Z_{obs} → compute p-value.

Reject H_0 if $p \leq \alpha$.

"Is this result surprising enough?"

Critical value approach

Compute Z_{obs} → compare to z_c .

Reject H_0 if **Z is past the threshold.**

"Is this result past the line?"

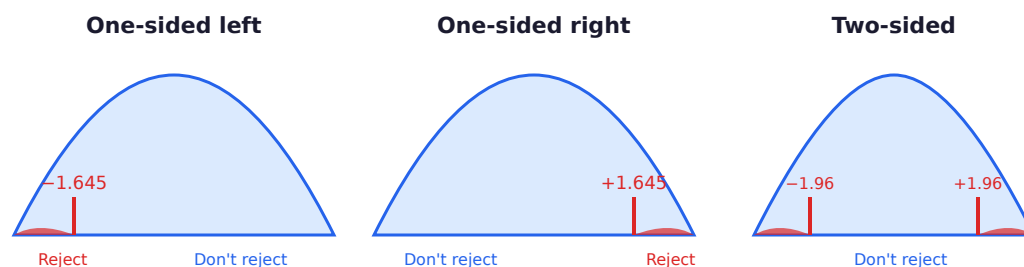
Same verdict, always. Both methods give identical reject/don't-reject decisions. The critical-value route is often faster on exams where α is fixed in advance.

4.2 Critical Values — The Threshold KEY

The **critical region** is the set of Z values that trigger rejection; its boundary is the **critical value z_c** . For $\alpha = 0.05$:

H_1	Test type	z_c	Reject H_0 if
$\mu < \mu_0$	Left	-1.645	$Z_{\text{obs}} \leq -1.645$
$\mu > \mu_0$	Right	+1.645	$Z_{\text{obs}} \geq +1.645$
$\mu \neq \mu_0$	Two-sided	± 1.96	$ Z_{\text{obs}} \geq 1.96$

Recognise these numbers? They're the exact z^* from the confidence intervals in Part 2 — no coincidence: both come from the same $N(0,1)$ and the same α . A two-sided test at α is literally "is μ_0 outside the $(1-\alpha)$ confidence interval?"



4.3 The Critical Region for $\bar{X}$ (not just Z) EXAM

You can express the threshold on the *sample-mean* scale directly — useful when a question asks "what value of $\bar{X}$ would trigger rejection?" Set $Z_{\text{obs}} = z_c$ in $Z = (\bar{X} - \mu_0)/\text{SE}$ and solve for $\bar{X}$:

CRITICAL VALUE FOR $\bar{X}$ (LEFT, $H_1: \mu < \mu_0$)

$$\bar{x}_c = \mu_0 - 1.645 \cdot \sigma/\sqrt{n}$$

CRITICAL VALUE FOR $\bar{X}$ (RIGHT, $H_1: \mu > \mu_0$)

$$\bar{x}_c = \mu_0 + 1.645 \cdot \sigma/\sqrt{n}$$

CRITICAL VALUES FOR $\bar{X}$ (TWO-SIDED)

$$\bar{x}_c = \mu_0 \pm 1.96 \cdot \sigma/\sqrt{n}$$

GPA example ($H_1: \mu < 4.4$, $\sigma = 0.538$, $n = 100$):

$$\text{SE} = 0.0538 \rightarrow \bar{x}_c = 4.4 - 1.645 \times 0.0538 = 4.4 - 0.0885 = \mathbf{4.3115}.$$

Reject H_0 if $\bar{X} \leq 4.3115$. Since $\bar{X} = 4.16 < 4.3115 \rightarrow$ reject. ✓ This is the literal "line in the sand" from Devlin's Chapter 4.

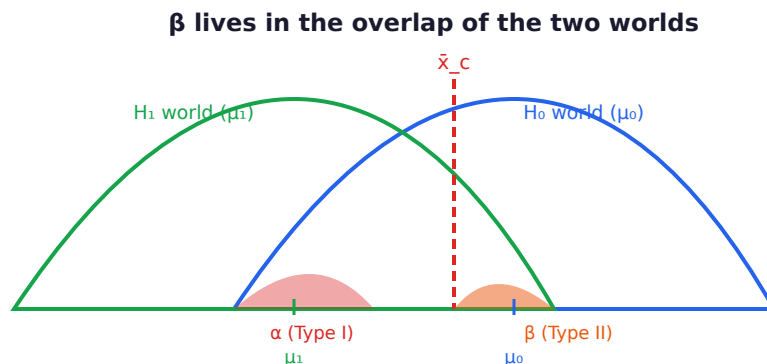
4.4 Type II Error — β KEY

The **Type II error** is failing to reject H_0 when H_1 is actually true — letting the guilty walk. Its probability is β . To compute it you need a **specific alternative μ_1** : the actual true mean if H_0 is wrong. (You can't measure "how often do I miss the effect?" without saying how big the effect really is.)

TYPE II ERROR PROBABILITY (LEFT TEST)

$$\beta(\mu_1) = P(\bar{X} > \bar{x}_c \mid \mu = \mu_1) = P(Z > (\bar{x}_c - \mu_1)/(\sigma/\sqrt{n}))$$

In words: if the truth is μ_1 , what's the chance $\bar{X}$ still lands in the "don't reject" zone and we miss it?



Reading it: blue = $\bar{X}$'s distribution if H_0 is true; green = if H_1 (μ_1) is true. Red (left of $\bar{x}_c$ under blue) = α . Orange (right of $\bar{x}_c$ under green) = β : the truth is μ_1 , but $\bar{X}$ drifted into the "don't reject" zone and we missed it. The more the two bells overlap, the bigger β .

4.5 Computing β — Step by Step EXAM

GPA: $H_0: \mu = 4.4$, $H_1: \mu < 4.4$, $\sigma = 0.538$, $n = 100$, $\alpha = 0.05$. Suppose the true mean is $\mu_1 = 4.2$. Find β .

- 1 Critical value (from H_0).** $\bar{x}_c = 4.4 - 1.645 \times 0.0538 = \mathbf{4.3115}$. Reject if $\bar{X} \leq 4.3115$.
- 2 $\beta = \mathbf{P}(\text{don't reject} \mid \mu = \mu_1)$.** $= P(\bar{X} > 4.3115 \mid \mu = 4.2)$.
- 3 Standardise using $\mu_1 = 4.2$** (not μ_0 !). $Z = (4.3115 - 4.2)/0.0538 = \mathbf{2.073}$.
- 4 Probability.** $\beta = P(Z > 2.073) = 1 - \text{NORM.DIST}(2.073; 0; 1; \text{TRUE}) = 1 - 0.9809 = \mathbf{0.0191} \approx \mathbf{1.9\%}$.

Interpretation: if the true mean really is 4.2, there's only a 1.9% chance the test fails to catch it. Sharp eyes.

The critical distinction: the test statistic standardises with μ_0 ; β standardises with μ_1 , the true alternative. Mixing them up is the classic β mistake.

4.6 Power of the Test KEY

POWER FOR ALTERNATIVE M_1

$$\text{Power}(\mu_1) = 1 - \beta(\mu_1)$$

Power is the probability of *correctly rejecting* H_0 when H_1 is true at μ_1 — the chance your test actually catches a real effect. It's the smoke detector with a live battery. High power = sensitive test.

Power	Meaning	Verdict
0.9	90% chance of detecting a real effect	Good
0.5	Coin flip	Poor
0.2	Misses 4 times in 5	Very poor

GPA continued: $\beta(4.2) = 0.019 \rightarrow \text{Power}(4.2) = 1 - 0.019 = \mathbf{0.981 = 98.1\%}$. If the true mean is 0.2 below the null, the test catches it 98% of the time. Excellent.

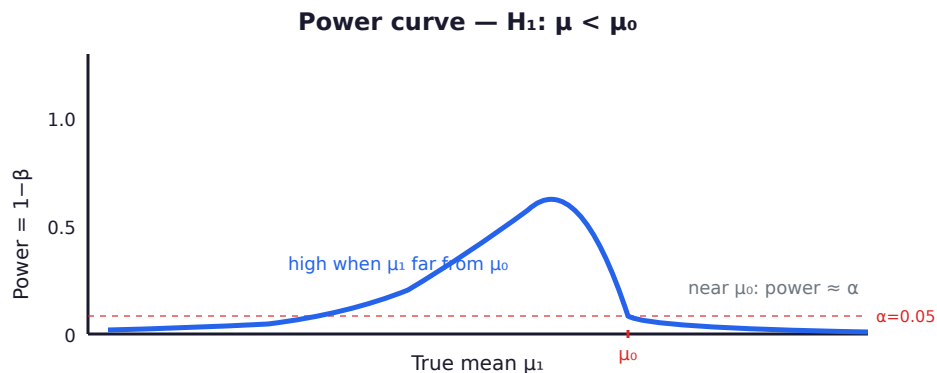
What raises power?

Factor	Power	Why
Larger n	↑	Smaller SE $\rightarrow$ the two bells overlap less $\rightarrow$ easier to tell apart
Larger $ \mu_1 - \mu_0 $	↑	Bigger true effect $\rightarrow$ more obvious departure
Smaller σ	↑	Less noise $\rightarrow$ the signal stands out
Larger α	↑	Wider rejection region $\rightarrow$ easier to reject (but more Type I errors)

The α - β trade-off (again): shrinking α raises β and lowers power — there's no free lunch. The *only* way to shrink both error types at once is to collect more data (raise n). That single lever, n , quietly runs through all four parts: it shrinks SE (P1), narrows CIs (P2), and now sharpens power (P4).

4.7 The Power Curve

Compute $\text{Power}(\mu_1)$ for every possible true value μ_1 and plot it:



- μ_1 **far from μ_0** (big true effect): Power ≈ 1 — almost certain to catch it.
- μ_1 **close to μ_0** : Power $\approx \alpha$ — a tiny effect is nearly invisible.
- $\mu_1 = \mu_0$ exactly: Power = α — that's just the Type I rate (there's nothing real to detect).

CHEAT SHEET

Master Quick Reference

Every formula in the chapter, in one place, in order.

Part 1 — Point estimators

Concept	Formula
Sampling distribution of $\bar{X}$	$\bar{X} \sim N(\mu, \sigma/\sqrt{n})$
Standard error	$SE = \sigma/\sqrt{n}$
Unbiased / consistent / efficient	$E(\bar{X}) = \mu \cdot SE \rightarrow 0 \text{ as } n \rightarrow \infty \cdot \text{smallest Var}$
σ vs SE	$\sigma = \text{spread of individuals; SE} = \text{spread of the mean}$

Part 2 — Confidence intervals

Concept	Formula / value
Two-sided CI (σ known)	$\bar{X} \pm z^* \cdot \sigma / \sqrt{n}$
Margin of error	$ME = z^* \cdot \sigma / \sqrt{n}$
z^* (two-sided)	90%: 1.645 · 95%: 1.96 · 99%: 2.576
z (one-sided 95%)	1.645
Required sample size	$n = (z^* \cdot \sigma / ME)^2 \rightarrow \text{round UP}$
Correct interpretation	"95% of such CIs contain μ " — NOT $P(\mu \text{ in CI})=0.95$

Part 3 — Hypothesis testing

Concept	Formula / rule
Test statistic	$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$
p-value (left / right)	$P(Z \leq z) / P(Z \geq z)$
p-value (two-sided)	$2 \times P(Z \geq z)$
Decision	reject H_0 if $p \leq \alpha$ (usually 0.05)
Type I / Type II	$\alpha = P(\text{reject} \mid H_0 \text{ true}) \cdot \beta = P(\text{not reject} \mid H_1 \text{ true})$
Language	"reject H_0 " (strong) / "do not reject H_0 " (weak) — never "accept H_0 "

Part 4 — Critical values & power

Concept	Formula
Critical value z_c ($\alpha=0.05$)	Left -1.645 · Right $+1.645$ · Two-sided ± 1.96

Critical value for $\bar{X}$
(left/right)

$$\bar{x}_c = \mu_0 \mp 1.645 \cdot \sigma / \sqrt{n}$$

β (left / right test)

$P(\bar{X} > \bar{x}_c \mid \mu_1) / P(\bar{X} < \bar{x}_c \mid \mu_1)$, standardise with μ_1

Power

$$1 - \beta(\mu_1)$$

Power rises with

larger n , larger $|\mu_1 - \mu_0|$, smaller σ , larger α

The one-sentence summary of the whole chapter.

You can only ever see a sample, so you use the sampling distribution $\bar{X} \sim N(\mu, \sigma / \sqrt{n})$ to (1) make a best guess, (2) wrap it in honest error bars, (3) decide whether a claim survives the evidence, and (4) check how good that decision procedure actually is. And the quantity $\sigma / \sqrt{n}$ — the standard error, the wobble — is the thread running through all four.

Merged & expanded from I1 Point Estimators · I2 Confidence Intervals · I3.1 Hypothesis Testing · I3.3 Critical Values & Power.